MCS3301 – Control Engineering I

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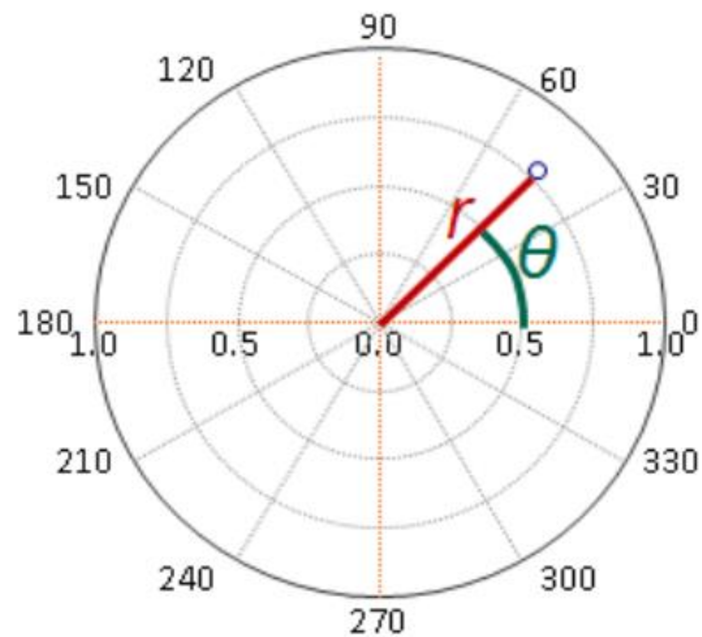
Polar Plots

- The stability of an LTI system can be determined graphically in the frequency domain by constructing a **polar** or **Nyquist** plot.
- The polar plot of a transfer function $G(j\omega)$ is a plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ on polar coordinates as ω is varied from zero to infinity.
- For an open loop transfer function $G(j\omega)H(j\omega)$, the polar form is
$$G(j\omega)H(j\omega) = |G(j\omega)H(j\omega)|\angle G(j\omega)H(j\omega)$$
- Thus, the polar plot is the locus of vectors as ω is varied from zero to infinity.
- In polar plots a positive (negative) phase angle is measured counterclockwise (clockwise) from the positive real axis.
- The **concentric circles** and the **radial lines** represent the magnitudes and phase angles respectively



Polar Plot: Construction

The following steps can be used to sketch a polar plot

i. Substitute, $s = j\omega$ in the open loop transfer function.

ii. Write the expressions for magnitude and the phase of $G(j\omega)H(j\omega)$

iii. Find the starting magnitude and the phase of $G(j\omega)H(j\omega)$ by substituting $\omega = 0$.

iv. Find the ending magnitude and the phase of $G(j\omega)H(j\omega)$ by substituting $\omega = \infty$.

v. Check whether the polar plot intersects the real axis, by making the imaginary term of $G(j\omega)H(j\omega)$ equal to zero and find the value(s) of ω.

vi. Check whether the polar plot intersects the imaginary axis, by making real term of $G(j\omega)H(j\omega)$ equal to zero and find the value(s) of ω.

vii. Other values of ω can be considered to sketch a more accurate polar plot.

Polar Plot: Example

Consider the open loop transfer function

$$G(s) = \frac{K}{s(\tau s + 1)}$$

$$G(j\omega = \frac{K}{j\omega(j\omega\tau + 1)} = \frac{K}{j\omega - \omega^2\tau}$$

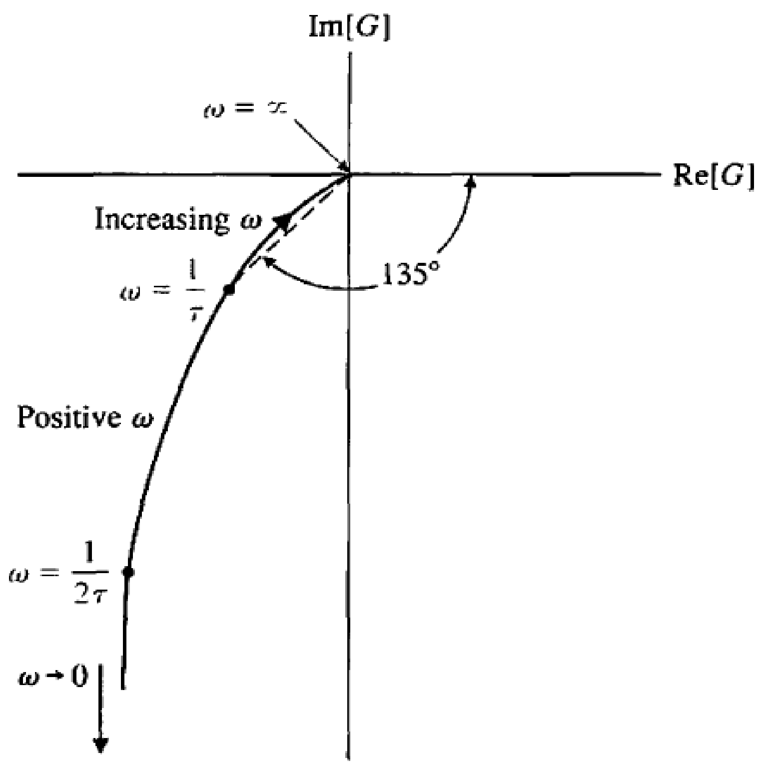
The magnitude of is

$$|G(j\omega)| = \frac{K}{\sqrt{\omega^2 + \omega^4\tau^2}}$$

And the phase as

$$\angle G(j\omega) = -\tan^{-1}\frac{1}{-\omega\tau}$$

ω	0	$1/2\tau$	$1/\tau$	∞
$\|G(j\omega)\|$	∞	$4K\tau/\sqrt{5}$	$K\tau/\sqrt{2}$	0
$\angle G(j\omega)$	-90	-117	-135	-180



Stability in the Frequency Domain

- Recall that the closed-loop transfer function of the system

$$M(s) = \frac{G(S)}{1 + G(S)H(S)}$$

- The characteristic equation is obtained by setting the denominator polynomial of $M(s)$to zero.
- Thus the roots of the characteristic equation are also the zeros of

$$1 + G(S)H(S) = F(s)$$

- To include multi-loop systems, the characteristic equation of a closed loop system can be expressed, as

$$F(s) = 1 + L(S) = 0$$

where $L(S) = \frac{N(s)}{D(s)}$

- Therefore the characteristic equation can be written as

$$F(s) = 1 + L(s) = 1 + \frac{N(s)}{D(s)} = \frac{D(s) + N(s)}{D(s)}$$

- It can be observed that
    i. Poles of the open loop transfer function $L(s)$ are the poles of $F(s)$
    ii. Zeros of $1 + L(s)$ are the poles of the closed-loop transfer function = roots of the characteristic equation.

Mapping Contours in the s-Plane

- Since stability requires all the zeros of $F(s)$ lie in the left-hand of the s-plane, Nyquist proposed a mapping of the right-hand of the s-plane into the $F(s)$-plane.
- A contour map is a contour or trajectory in one plane mapped or translated into another plane by a relation $F(s)$.
- Since s is a complex variable $(s = \sigma + j\omega)$, the function $F(s)$ is also complex.
- It can be defined as $F(s) = u + jv$ and thus represented on a complex $F(s)$-plane with coordinates u and v.
- Consider a function $F(s) = 2s + 1$
- A close contour Γ_s is arbitrarily chosen in the s-plane, which does not pass through any pole or zero of $F(s)$.
- Let the contour Γ_s be defined by the points $A(1, j1), B(1, -j1), C(-1, -j1)$ and $D(-1, j1)$.
- The s-plane can thus be mapped to the $F(s)$-plane through the relation $F(s)$.

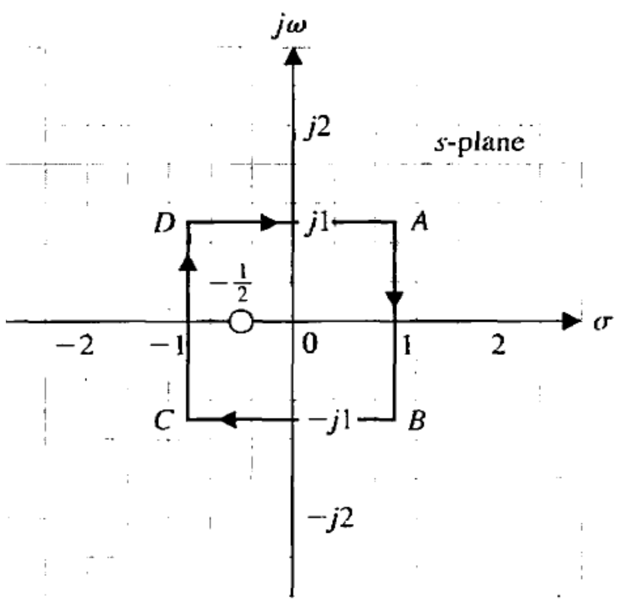
Mapping Contours in the s-Plane

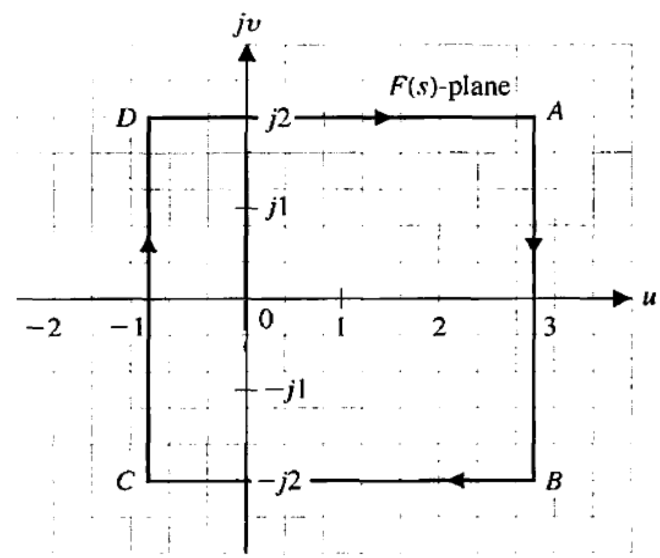
- Therefore

$$u + jv = F(s) = 2s + 1 = 2(\sigma + j\omega) + 1$$

- Thus for this specific case $u = 2\sigma + 1$ and $v = 2\omega$.
- The points $ABCD$ in the Γ_s thus respectively map onto the points $ABCD$ on the F(s) plane.

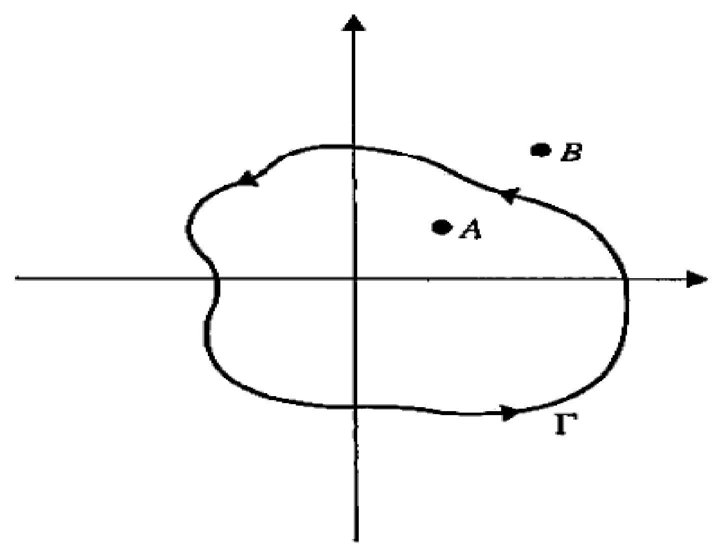
	A	B	C	D
$\Gamma_s = \sigma + j\omega$	$1 + j1$	$1 - j1$	$-1 - j1$	$-1 + j1$
$F(s) = u + jv$	$3 + j2$	$3 - j2$	$-1 - j2$	$-1 + j2$





Mapping Contours: Encirclements

- Encircled: A point or region in a complex function plane is said to be encircled by a closed path if it is found inside the path.
- For example, point A in the diagram is encircled by the closed path Γ, because A is inside the closed path.
- Point B is not encircled by the closed path Γ, because it is outside the path.
- When the closed path Γ has a direction assigned to it, the encirclement can be in the clockwise (CW) or the counterclockwise (CCW) direction.
- As shown, point A is encircled by Γ in the CCW direction.
- We can say that the region inside the path is encircled in the prescribed direction, and the region outside the path is not encircled.

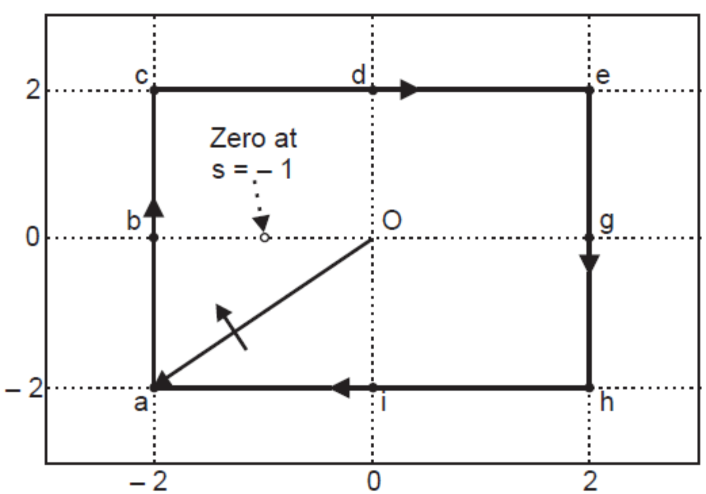


Mapping Contours in the s-Plane

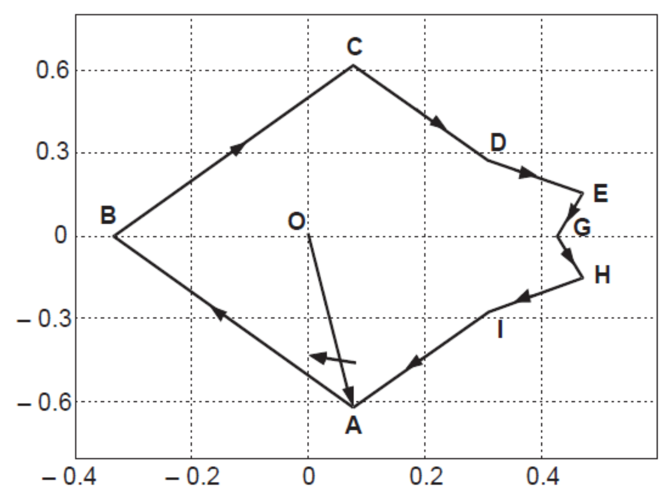
- Consider the rational function $F(s) = \frac{s+1}{s+5}$
- Take the closed contour points on the s-plane $a(-\ 2 - \ j2), b(-2\ +\ j0)$, $c(-\ 2\ +\ j2), d(0\ +\ j2), e(2\ +\ j2)$, $g(2\ + j0), h(2 -\ j2)$ and $i(0 -\ j2)$.
- The corresponding points on the $F(s)$-plane are calculated as shown in the Table

$\Gamma_s = \sigma + j\omega$	$F(s) = u + jv$
$a(-2 - j2)$	A$(0.0769 - j0.6154)$
$b(-2 + j0)$	B$(-0.3333 + j0.0)$
$c(-2 + j2)$	C$(0.0769 + j0.6154)$
$d(0 + j2)$	D$(0.3103 + j0.2759)$
$e(2 + j2)$	E$(0.4717 + j0.1509)$
$g(2 + j0)$	G$(0.4286 + j0)$
$h(2 - j2)$	H$(0.4717 - j0.1509)$
$i(0 - j2)$	I$(0.3103 - j0.2759)$

$\boldsymbol{\Gamma_s}$



$\boldsymbol{F(s)}$



Mapping Contours in the s-Plane

The following points are noteworthy

i. The mapping of a **close** contour in the s-plane results in a **close** contour in the $F(s)$-plane, when the chosen close contour in the s-plane does not pass through any poles and zeroes of the function $F(s)$.

ii. The origin $(0,0)$ is inside the close contour *abcdeghi* in the s-plane. The origin $(0,0)$ in the $F(s)$ plane is also found to be inside the closed contour *ABCDEGHI* in the $F(s)$ plane for the chosen closed contour *abcdeghi* in the s-plane.

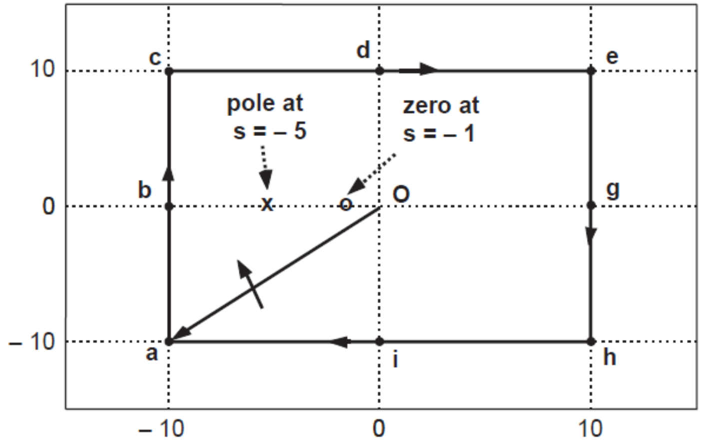
iii. The zero at s = – 1 is inside the closed contour *abcdeghi* in the s-plane whereas the pole at s = – 5 is not.

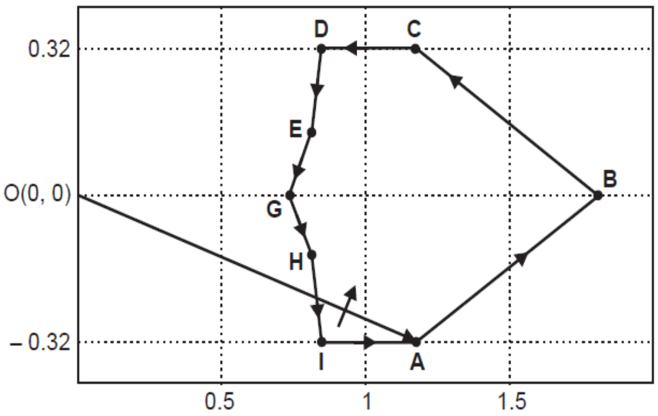
iv. As we traverse along the contour *abcdeghi* in the clockwise direction, the traversal in the contour ABCDEGHI in the $F(s)$-plane is also in the clockwise direction.

v. By convention, the area within a contour to the right of the traversal of the contour is considered to be the area **enclosed** by the contour.

Mapping Contours in the s-Plane

- Consider the mapping of the same function $F(s)$ for different closed contour *abcdeghi* in the s-plane such that both the zero at s = – 1 and the pole at s = – 5 of the function $F(s)$ are enclosed.
- The corresponding mapping in the F(s)-plane is again a closed contour but the direction of traversal of the contour in the F(s)-plane is anticlockwise even though the direction of traversal in the s-plane is clockwise.
- This is because the contour *abcdeghi* in the s-plane encloses both the zero at s = – 1 and the pole s = – 5.
- According to the convention of enclosure, the area covered by ABCDEGHI in the $F(s)$-plane is not "enclosed".





Principle of the Argument

- The encirclement of the origin by the closed contour Γ_F in the F(s)-plane can be related to the encirclement of the origin by the closed path Γ_s in the s-plane with the help of **Cauchy's theorem**.
- Cauchy's theorem known as the **principle of the argument** is stated as:

> If a contour $\boldsymbol{\Gamma_s}$ in the s-plane encircles ***Z*** zeros and ***P*** poles of ***F(s)*** and does not pass through any poles or zeros of ***F(s)*** and the traversal is in the **clockwise** direction along the contour, the corresponding contour $\boldsymbol{\Gamma_F}$ in the F(s)-plane encircles the origin of the F(s)-plane $N = Z - P$ times in the **clockwise** direction.

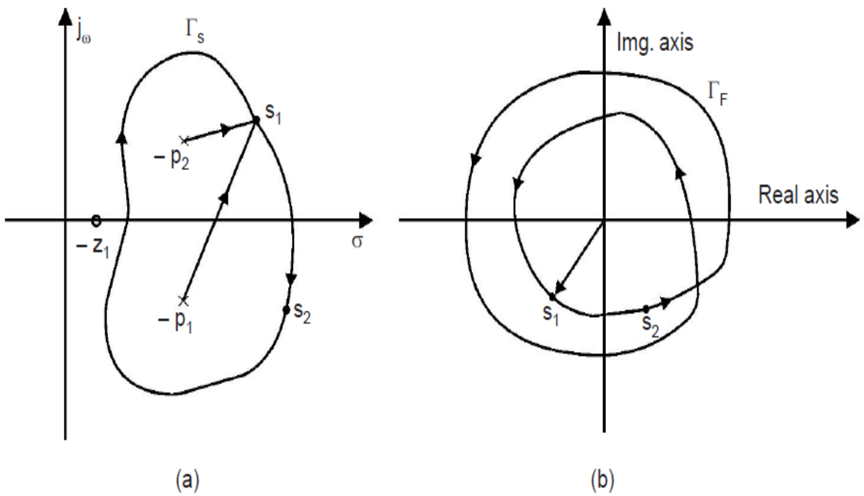
- Re-examining the previous examples it is observed that

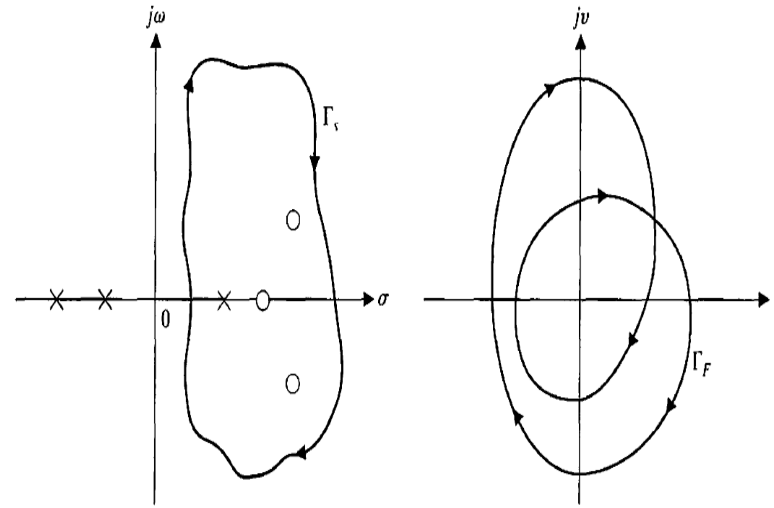
$N = 1 - 0 = 1$ hence 1 encirclement of the origin of *Fs* (Slide 7)

$N = 1 - 0 = 1$ hence 1 encirclement of the origin of *Fs* (Slide 9)

$N = 1 - 1 = 0$ hence 0 encirclements of the origin of *Fs* (Slide 11)

Principle of the Argument: Illustration





- The contour $\boldsymbol{\Gamma_s}$ encloses no zeros and two poles
- Thus in this case $N = 0 - 2 = -2$
- It is observed that the contour $\boldsymbol{\Gamma_F}$ completes **two CCW** encirclements of the origin in the F(s)-plane

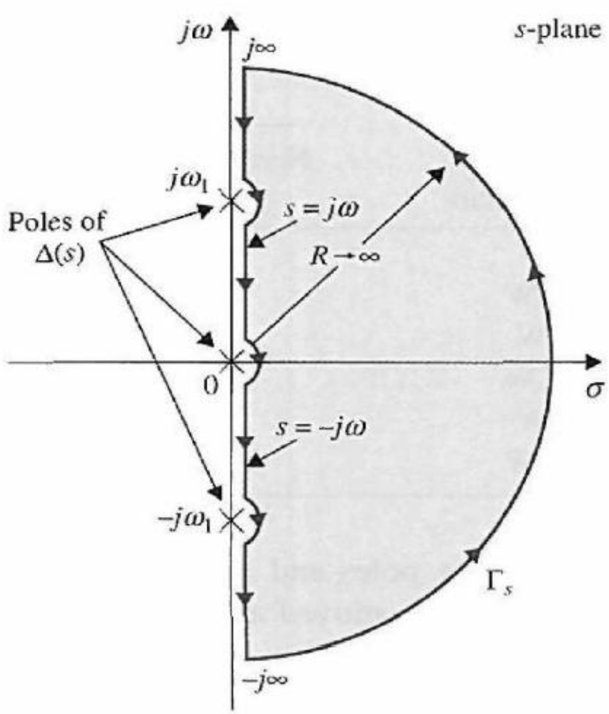
- The contour $\boldsymbol{\Gamma_s}$ encloses three zeros and one pole
- Therefore $N = 3 - 1 = 2$
- The contour $\boldsymbol{\Gamma_F}$ completes **two CW** encirclements of the origin in the F(s)-plane

The Nyquist Stability Criterion

- The Nyquist criterion is a direct application of the principle of the argument when the *s*-plane locus $\boldsymbol{\Gamma_s}$ is taken to encircle the entire right of the s-plane known as the **Nyquist path**.
- Since the Nyquist path must not pass through any poles and zeros of $F(s)$, the small semicircles shown along the $j\omega$ -axis in to indicate that the path should go around these poles and zeros if they fall on the $j\omega$ -axis

- The stability of a closed-loop system can be determined by plotting the $F(s) = 1 + L(s)$ locus when *s* takes on values along the Nyquist path
- The behavior of the $F(s)$ plot is then and investigated with respect to the critical point (origin of the F(s)-plane in this case).
- More conveniently, a function $F'(s)$ may be defined as

$$F'(s) = F(s) - 1 = L(s)$$



The Nyquist Stability Criterion

- In this case, the number of clockwise encirclements of the origin of the F(s)-plane becomes the number of clockwise encirclements of the -1 point in the $F'(s) = L(s)$-plane
- Hence the $(-\mathbf{1}, \boldsymbol{j}\mathbf{0})$ point in the $L(s)$-plane becomes the **critical point** for the determination of **closed-loop stability**.
- For single-loop systems, $L(s) = G(s)H(s)$ and the closed-loop stability can be determined by investigating the behavior of the $G(s)H(s)$ plot with respect to the $(-1, j0)$ point of the $G(s)H(s)$-plane
- Thus, given a control system that has the characteristic equation $1 + L(s) = 0$, the Nyquist criterion involves the following steps
  i. The L(s) plot corresponding to the Nyquist path is constructed in the L(s)-plane.
  ii. The value of N, the number of encirclements of the $(-1, j0)$ point made by the L(s) plot, is observed.
  iii. 4. The Nyquist criterion follows from $N = Z - P$.

The Nyquist Stability Criterion

- The Nyquist stability criterion can be thus be stated as

> A feedback control system is stable if and only if, for the contour $\boldsymbol{\Gamma_L}$, the number of counterclockwise encirclements of the $(-1, \mathrm{j}0)$ point is equal to the number of poles of $L(s)$ with positive real parts.

- Thus for systems with no open loop poles on the right of the s-plane, the Nyquist stability criterion can be states as

> A feedback system is stable if and only if the contour $\boldsymbol{\Gamma_L}$ in the L(s)-plane does not encircle the $(-1, j0)$ point when the number of poles of *L(s)* in the right-hand s-plane is zero (*P* = 0).

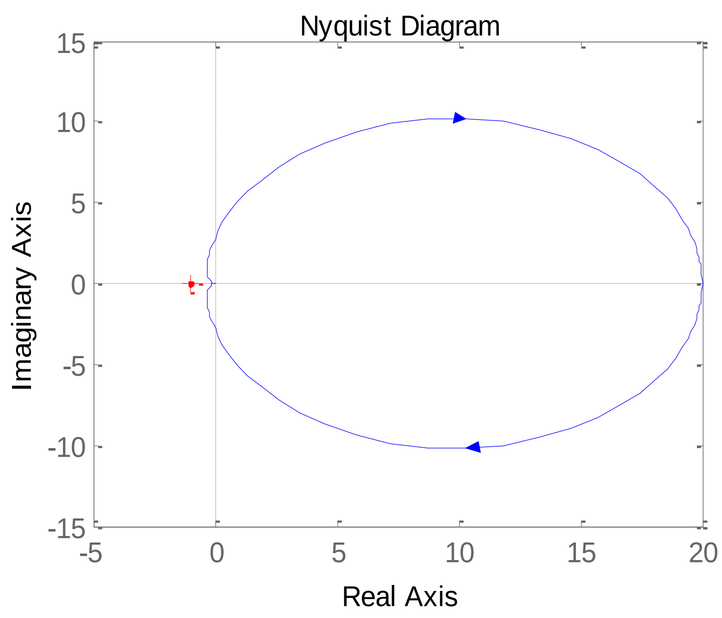
Nyquist Criterion: Examples

Example 1: Using the Nyquist criterion, comment on the stability of the system with the transfer function

$$L(s) = G(s)H(s) = \frac{10}{(s+0.1)(s+5)}$$

- From the poles of $L(s)$, P = 0
- The Nyquist path does not encircle the critical point $(-1, j0)$.
- Therefore $Z = N + P = 0 + 0 = 0$ implying none of the zeros of the CE $1 + G(s)H(s) = 0$ is on the RHS of the imaginary axis in the s-plane.
- Thus according to the Nyquist criterion the system is stable.
- The commands below plot the Nyquist diagram in MATLAB

```
>> num = 10;
>> den = conv([1 0.1], [1 5]);
>> nyquist(num, den)
```



Nyquist path of $L(s)$